

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_stratified_random_replay.
- Best TSPLIB holdout gap: phase6_stratified_random_replay.
- Best synthetic holdout gap: phase6_failure_replay.

Run Metadata

- run_name: run_20260519_005032_h
- started_at_local: 2026-05-19 00:50:32
- finished_at_local: 2026-05-19 01:09:42
- duration_hhmm: 00:19
- duration_seconds: 1150.704
- seed_offset: 7000
- replicate_label: h
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.0	0.842525	0.63	-0.02956
phase6_random_replay	random	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.19457	0.864534	0.56	-0.028807
phase6_failure_replay	failure	score_only	False	0.105503	0.0	0.067138	0.171206	0.249441	0.199982	0.0	0.76	0.0

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.213387	0.064916	0.159398	0.236772	0.214847	0.19457	0.0	0.56	0.0
phase6_stratified_random_replay	stratified_random	score_only	False	0.08682	0.003941	0.056682	0.236772	0.148504	0.221162	0.788169	0.76	0.04152
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.126568	0.0	0.080543	0.236772	0.187137	0.157119	0.759221	0.76	0.080585
phase6_residual_failure_replay	residual_failure	score_only	False	0.117653	0.010343	0.078631	0.101234	0.27062	0.373989	0.0	0.76	0.0
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.163475	0.0	0.10403	0.171206	0.247432	0.17767	0.0	0.76	0.0
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.204223	0.0	0.12996	0.171206	0.247952	0.175969	0.0	0.76	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.842525, and final complexity 0.64.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency -0.02956 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.399069, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": 0.002249}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.133545}],

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{"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.321274, "family": "lin", "name": "lin105",  
"optimality_gap": 0.328813, "residual_gap": 0.007539}]
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phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.864534, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133486, mean selected residual gap 0.03788.
- Adaptation efficiency -0.028807 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.402714, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.001396}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.134817}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.317296, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.011517}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.105503, synthetic 0.0, combined 0.067138.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.249441, size bias 0.120322, and failure concentration 0.199982.
- Replay selection diversity 0.171194, mean selected expected gap 0.291967, mean selected residual gap -0.004933.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.105503 across 7 instances; family means: ch=0.074679, kroD=0.081948, pcb=0.226476, pr=0.167466, rd=0.065866, st=0.047407.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 7542, "cost": 10449, "expected_gap": 0.245107, "family": "berlin", "name": "berlin52", "optimality_gap": 0.385442, "residual_gap": 0.140335}, {"best_known_cost": 2579, "cost": 3390, "expected_gap": 0.404808, "family": "a", "name": "a280", "optimality_gap": 0.314463, "residual_gap": -0.090345}, {"best_known_cost": 14379, "cost": 17495, "expected_gap": 0.328729, "family": "lin", "name": "lin105", "optimality_gap": 0.216705, "residual_gap": -0.112024}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132655, mean selected residual gap 0.038711.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.400776, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": 0.000542}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.137679}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.328131, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.000682}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.08682, synthetic 0.003941, combined 0.056682.
- Accepted-epoch count 3, mean accepted code novelty 0.788169, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.148504, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.238172, mean selected residual gap -0.05168.
- Adaptation efficiency 0.04152 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.08682 across 7 instances; family means: ch=0.090348, kroD=0.092514, pcb=0.167179, pr=0.036113, rd=0.052718, st=0.078519.
- Panel synthetic_holdout mean gap 0.003941 across 4 instances; family means: clustered_gaussian=0.015764, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3477, "expected_gap": 0.40031, "family": "a", "name": "a280", "optimality_gap": 0.348197, "residual_gap": -0.052113}, {"best_known_cost": 14379, "cost": 17054, "expected_gap": 0.329634, "family": "lin", "name": "lin105", "optimality_gap": 0.186035, "residual_gap": -0.143599}, {"best_known_cost": 7542, "cost": 8692, "expected_gap": 0.245028, "family": "berlin", "name": "berlin52", "optimality_gap": 0.152479, "residual_gap": -0.092549}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.126568, synthetic 0.0, combined 0.080543.
- Accepted-epoch count 3, mean accepted code novelty 0.759221, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.187137, size bias 8.3e-05, and failure concentration 0.157119.
- Replay selection diversity 0.272437, mean selected expected gap 0.156895, mean selected residual gap 0.009934.

- Adaptation efficiency 0.080585 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.126568 across 7 instances; family means: ch=0.076935, kroD=0.082605, pcb=0.322384, pr=0.072236, rd=0.176359, st=0.078519.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3376, "expected_gap": 0.400931, "family": "a", "name": "a280", "optimality_gap": 0.309035, "residual_gap": -0.091896}, {"best_known_cost": 7542, "cost": 8379, "expected_gap": 0.240918, "family": "berlin", "name": "berlin52", "optimality_gap": 0.110979, "residual_gap": -0.129939}, {"best_known_cost": 629, "cost": 667, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.060413, "residual_gap": -0.193641}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.117653, synthetic 0.010343, combined 0.078631.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.101234, mean hardness 0.27062, size bias 0.065785, and failure concentration 0.373989.
- Replay selection diversity 0.123685, mean selected expected gap 0.268811, mean selected residual gap 0.03726.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.117653 across 7 instances; family means: ch=0.068952, kroD=0.184794, pcb=0.201406, pr=0.114877, rd=0.095702, st=0.088889.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3430, "expected_gap": 0.402869, "family": "a", "name": "a280", "optimality_gap": 0.329973, "residual_gap": -0.072896}, {"best_known_cost": 629, "cost": 835, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.327504, "residual_gap": 0.065501}, {"best_known_cost": 14379, "cost": 18695, "expected_gap": 0.322164, "family": "lin", "name": "lin105", "optimality_gap": 0.30016, "residual_gap": -0.022004}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.163475, synthetic 0.0, combined 0.10403.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.247432, size bias 0.120322, and failure concentration 0.17767.
- Replay selection diversity 0.201519, mean selected expected gap 0.316557, mean selected residual gap -0.043209.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.163475 across 7 instances; family means: ch=0.101907, kroD=0.159998, pcb=0.175529, pr=0.334443, rd=0.223135, st=0.047407.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3108, "expected_gap": 0.403257, "family": "a", "name": "a280", "optimality_gap": 0.205118, "residual_gap": -0.198139}, {"best_known_cost": 629, "cost": 678, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.077901, "residual_gap": -0.18919}, {"best_known_cost": 7542, "cost": 8123, "expected_gap": 0.233148, "family": "berlin", "name": "berlin52", "optimality_gap": 0.077035, "residual_gap": -0.156113}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.204223, synthetic 0.0, combined 0.12996.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.247952, size bias 0.120322, and failure concentration 0.175969.
- Replay selection diversity 0.195253, mean selected expected gap 0.310443, mean selected residual gap -0.04729.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.204223 across 7 instances; family means: ch=0.275139, kroD=0.358223, pcb=0.24162, pr=0.111142, rd=0.14311, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19792, "expected_gap": 0.328841, "family": "lin", "name": "lin105", "optimality_gap": 0.376452, "residual_gap": 0.047611}, {"best_known_cost": 2579, "cost": 3450, "expected_gap": 0.40349, "family": "a", "name": "a280", "optimality_gap": 0.337728, "residual_gap": -0.065762}, {"best_known_cost": 7542, "cost": 9044, "expected_gap": 0.237364, "family": "berlin", "name": "berlin52", "optimality_gap": 0.199151, "residual_gap": -0.038213}]

Judge Appendix

Summary of Key Findings on Replay-Aware TSP Benchmark Suite

1. Transfer Performance (Primary focus on Held-out TSPLIB gap and Synthetic Holdout gap)

Condition	Mean Held-out TSPLIB Gap	Mean Synthetic Holdout Gap	Mean Transfer Gap
phase6_stratified_random_replay	**0.08682**	**0.003941**	**0.056682**
phase6_failure_replay	0.105503	0.0	0.067138
phase6_diversity_weighted_replay	0.126568	0.0	0.080543
phase6_residual_failure_replay	0.117653	0.010343	0.078631
phase6_diversity_failure_replay	0.163475	0.0	0.10403
phase6_random_replay_compression	0.213387	0.064916	0.159398
phase6_random_replay	0.213387	0.064916	0.159398
phase6_no_replay	0.213387	0.064916	0.159398
phase6_diversity_failure_replay_compression	0.204223	0.0	0.12996

- ****Best overall transfer performance:**** phase6_stratified_random_replay (lowest held-out TSPLIB gap, synthetic holdout gap, and transfer gap).
- phase6_failure_replay has the best synthetic holdout gap (0.0) and competitive held-out TSPLIB and transfer gaps, indicating good transfer on synthetic but slightly worse on real TSPLIB.
- Conditions with replay compression generally have higher gaps, indicating compression may negatively affect transfer.

2. Adaptation Efficiency and Archive Characteristics

Condition	Adaptation Efficiency	Archive Descriptor Diversity	Archive Hardness	Archive Size Bias	Replay Failure Concentration
phase6_stratified_random_replay	**0.04152**	0.236772	0.175619	0.0	0.221162
phase6_diversity_weighted_replay	0.080585	0.240179	0.196328	0.0	0.342222
phase6_failure_replay	0.0	0.174718	0.320013	0.124979	0.23356
phase6_residual_failure_replay	0.0	0.149653	0.318916	0.094391	0.188209
phase6_diversity_failure_replay	0.0	0.174718	0.295958	0.124979	0.174603
phase6_random_replay	-0.028807	0.236772	0.214847	8.3e-05	0.19457
phase6_random_replay_compression	0.0	0.236772	0.212361	0.0	0.16
phase6_no_replay	-0.02956	0.240179	0.212361	0.0	0.0
phase6_diversity_failure_replay_compression	0.0	0.174718	0.300947	0.124979	0.175969

- Replay conditions with failure-based archives (failure, residual_failure, diversity_failure) show ****higher hardness (~0.3) and moderate diversity (0.15-0.17)**** than non-failure ones.
- phase6_diversity_weighted_replay shows highest adaptation efficiency (0.080585) but worse transfer gap than stratified random replay.
- phase6_stratified_random_replay balances moderate diversity and lower hardness with improved transfer.

3. Replay Mechanism Effects

- ****Broad-Coverage Replay (stratified_random):****
- Achieves best transfer performance (lowest TSPLIB and synthetic gaps).
- Archive has moderate diversity and lowest hardness, indicating balanced coverage.
- Failure concentration is moderate (~22%), showing replay is not overly focused on failures.
- Selection mode is score_only, possibly enabling effective sample prioritization.
- ****Raw Failure Replay (failure, diversity_failure):****
- Archive hardness is highest (≥ 0.29), size bias present (~0.12).

- Final synthetic gaps are zero (perfect on synthetic holdout), but transfer gaps are higher compared to stratified_random replay.
- Replay failure concentration ranges 17-23%, moderate to high.
- Code novelty is null (0.0), indicating no new code is added during these replays.
- **Residual Failure Replay (residual_failure):**
- Moderate archive hardness (~0.32) and diversity (lowest, 0.15).
- Transfer and TSPLIB gaps slightly better than raw failure replay but worse than stratified random.
- Shows compression aware behavior with moderate replay failure concentration (~18%).
- Code novelty is zero, similar to failure replay.
- **Diversity-Weighted Replay (diversity_weighted):**
- Highest adaptation efficiency (0.080585) suggesting effective learning.
- Archive diversity high (0.24) with moderate hardness (0.19).
- Transfer gaps intermediate (0.08).
- Replay failure concentration highest (~34%), suggests more focused on difficult or rare failures.
- Code novelty is zero, indicating no novel code discovered despite better efficiency.
- **Replay with Compression:**
- Appears in random and diversity-failure replay with lower code novelty (0.0).
- Generally associated with higher final transfer gaps (~0.13-0.16) and TSPLIB gaps (~0.20+).
- Compression tends to reduce archive diversity slightly without clear benefits in transfer.

4. Code Novelty vs Transfer

- The phase6_failure_replay (best synthetic condition) has **zero code novelty but improved synthetic gap (0.0)** and improved transfer gap (0.067).
- phase6_stratified_random_replay (best holdout and transfer condition) retains **substantial code novelty (~0.76-0.88)** with best transfer gaps.
- Failure-based replays tend to have code novelty at zero while stratified_random replay encourages code novelty without compromising transfer.
- Therefore, code novelty can decline while transfer improves (failure-replay conditions).
- Lexical novelty noted does not imply algorithmic invention without matching transfer gains.

5. Failure Concentration and Replay Diversity

- Failure-based replay conditions (failure, residual_failure, diversity_failure) exhibit higher failure concentration (~17-37%) vs none or minimal for other types.
- Diversity-weighted replay increases failure concentration most (~34%), possibly targeting varied difficult cases.

- Stratified random replay balances failure concentration (~22%) and diversity (~0.24), correlating with best transfer.
- Raw random and no-replay conditions show zero or negligible failure concentration indicating less focused replay.

****Summary and Recommendations****

- ****Best Transfer Strategy:**** phase6_stratified_random_replay achieves the lowest held-out TSPLIB and synthetic gaps, with moderate replay failure concentration and high archive diversity, indicating broad effective replay.
- ****Best Synthetic Transfer:**** phase6_failure_replay yields perfect synthetic holdout scores but at the cost of somewhat reduced transfer on TSPLIB instances compared to stratified random.
- ****Code Novelty Decoupled from Transfer:**** Conditions with zero code novelty (failure replays) can still improve transfer, indicating that replay-driven sampling of hard cases, rather than new code generation, drives performance gains.
- ****Replay Compression Impact:**** Replay compression reduces archive diversity and fails to improve transfer performance relative to non-compressed variants.
- ****Failure Replay Variants:**** Failure and residual failure replays bias towards harder cases and have higher failure concentration but do not necessarily lead to best transfer gaps.
- ****Recommendation:**** Prioritize broad-coverage and stratified random replay strategies for generalizability and transfer, balancing archive diversity and failure focus, while maintaining some code novelty. Avoid compression-induced replay constraints that may degrade performance.

Interpretation aligned strictly with indicated metrics, avoiding narrative speculations.